

Self Learning of Web Programming Technology for PGCP-AC

JS and DOM : (6 hr)

1. Explore the JSON syntax and create JSON Object and JSON Array of Books
2. Explore the class syntax and perform association and inheritance with example
3. Create an html page that provides the input for book details (id, name, author, price, booktype) and display those in the table

Tailwind : (2 hrs)

1. Add the tailwind to your working directory by downloading the files. Add in your html page and create a simple login and registration page.
2. Explore the flex from the tailwind
3. Display dummy data of products on html page using cards from daisyUI in flex display
4. Use the navbar from daisyUI and customize it with different colors and links in it.also make it work for different screen size

AJAX& Axios HTTP Client (2 hrs)

1. Create an web application that displays the book records from server
 - using XHR
 - fetch using then..catch
 - fetch using async-await
 - using axios

TypeScript : (2 hrs)

1. Handle the API calls using typed promises for the dummy data of users from dummyjson website. display the data in the table in html page

Node : (4 hrs)

1. Search for Event Loop Visualizer and understand the behavior of Event Loop.
2. Create a simple node application that performs add,sub,mul,sub by taking user input from console. Explore how to take input from console in node js
3. Create a simple http server and handle the api routes for products to perform GET,POST,PUT,DELETE. Perform the CRUD operation in a simple array with some dummy data
4. Explore the different ways of importing and exporting the modules in Node JS specifically the Common JS and Modern JS

Express : (2 hrs)

1. Create your own Express Server that have the API created for Courses and Students. For this have the course table and the student table. Perform all the CRUD operation on both the tables. Test the api's for

both table on postman.

React : (3 hrs)

1. Note down the difference between the MPA and SPA and the examples when to use which one
2. Explore the virtual dom and what is diffing
3. Create a react application using a built tool (eg-vite). Understand the Project structure and note down their description
4. Explore the new syntax of JSX and note the syntax change to use html tags

React- Props : (3 hrs)

1. Explore what is props drilling and the solution to it. You can create an example with props drilling for better understanding

React Router : (3 hrs)

1. Create a react application which will have multiple components which will be loaded using react router. Implement the static and dynamic navigation for the same.

React-Redux : (3 hrs)

1. Understand the entire working of redux and implement the same.
2. Differentiate between the context and redux and when to use which one, note it down.